

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR(Computer Science)
SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours

Batch 2023

Dated : 05-JUN-26

Max Marks : 60

Web Engineering - CT-362

Q1	CL0 3	Marks 20
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A private healthcare company wants to digitize its clinic operations because patients currently face long waiting times, lost prescriptions, and poor appointment management. They have hired your software development team to build a complete mobile application solution.

You are assigned as a **Full Stack Website Developer** to design and develop a real-world application using:

- **Frontend:** React Js
- **Backend:** Node.js + Express.js
- **Database:** Of Your choice

The clinic has multiple doctors and receives hundreds of patients daily. The management wants a website where:

- Patients can register and book appointments.
- Doctors can manage schedules and prescriptions.
- Admin can monitor clinic activities and reports.
- Notifications are sent for upcoming appointments.
- Patients can upload medical reports and prescriptions.

The system must support secure login, real-time appointment status, and role-based access.

Q2	CLO 4	Marks 20
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A software company wants to analyze employee performance records using modern JavaScript functionalities. The company stores employee data in JSON format and wants developers to perform multiple operations using advanced JavaScript concepts such as:

1) map() 2) filter() 3) forEach() 4) find() 5) sort() 6) async/await 7) Promises 8) destructuring 9) spread operator

```
const employees = [  
  { id: 1, name: "Ali Khan", department: "Development", salary: 95000, experience: 4, tasksCompleted: 45, rating: 4.5, isActive: true },  
  
  { id: 2, name: "Sara Ahmed", department: "Design", salary: 85000, experience: 3, tasksCompleted: 38, rating: 4.2, isActive: true },
```

P.T.O

```
{ id: 3, name: "Usman Tariq", department: "Development", salary: 120000, experience: 6, tasksCompleted: 60, rating: 4.9, isActive: false },
```

```
{ id: 4, name: "Ayesha Noor", department: "Marketing", salary: 70000, experience: 2, tasksCompleted: 25, rating: 3.8, isActive: true },
```

```
{ id: 5, name: "Hamza Ali", department: "HR", salary: 65000, experience: 5, tasksCompleted: 40, rating: 4.1, isActive: true }
];
```

- Use `forEach()` to display: 1) Employee Name 2) Department
- Use `map()` to create: Employee Name - Salary
- Use `filter()` to find: Salary > 80,000 Rating > 4.0 Active Employees
- Use `find()` to search: Employee with ID = 3
- Use `reduce()` to calculate: Total Salary, Average Rating
- Use `Sort()` to Salary (High → Low) Experience (Low → High)
- Use Destructuring: Extract: name, salary, department
- Use Spread Operator: Add new employee, Clone employee array
- Create a Promise: Return employee data after 3 seconds
- Use `async/await`: Fetch and display employee data
- Use `try-catch`: Handle errors properly
- Create bonus function: Give 10% bonus to employees with rating > 4.5
- Generate Performance Report: Name | Tasks Completed | Status
- Conditions: Above 50 → Excellent 30-50 → Good Below 30 → Needs Improvement

Q3	CLO 2	Marks 10
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Draw a complete system architecture diagram of the scenario discussed in Q1.

Note: You have to show a separate diagrams for a complete architecture, Backend and Databases. Show all necessary details required.

Q4	CLO 2	Marks 10
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A company is developing an E-Commerce Website where different users interact with the system.

There are three types of users:

- 1) Customer
- 2) Seller
- 3) Admin

The company wants to secure the application because users perform online payments, manage products, and access personal information.

System Requirements:

- Users must register and log in securely.
- Passwords should never be stored in plain text.
- Customers can purchase products only.
- Sellers can add and manage products.
- Admin can manage the complete platform.
- Unauthorized access must be restricted.

As a full stack developer, explain how Authentication and Authorization will be implemented in this application

1. React JS Frontend	2. Node.js + Express Backend	3. Database	4. bcrypt	5. JWT (JSON Web Token)
Login/Register Screens	Handle Login APIs	Store User Data	Hash Passwords during registration	Maintain login session
Form Validation	Verify Users	Store Hashed Passwords	Compare Passwords during login	Authenticate API requests
Store JWT Token	Generate JWT Tokens	Store Roles and Permissions		Secure user identity
API Integration	Protect Private Routes			